5. So by contradiction ⇒ L is not a TL CFG

(85) Push down Automata 5.7

A push down Automata (PDA) is a way to implement a Context Free Grammar in a similar way we design Finite Automata for Regular Grammar

→ It is More powerful than FSM

→ FSM has a very limited memory but PDA has more memory

→ PDA = Finite State Machine + A Stack

↑ infinite memory

Stack: PUSH : A new element add to Top

pop : The Top element of stack removed

A Push down Automata has 3 components:

1) An input tape

2) A Finite Control Unit

3) A Stack with infinite Size

(86) PDA Formal Definition

FNA: $(Q, \Sigma, q_0, F, \delta)$

$P = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$

where:

Q = A Finite set of States

Σ = A finite set of Input Symbols

Γ = A finite Stack Alphabet

δ = The Transition Function

q_0 = The start State

z_0 = The Start stack Symbol

F = The set of Final / Accepting States

$\delta(q, a, X)$ where

i) q is a state in Q ii) a is either an

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Input Symbor in Σ or $a = \epsilon$

(iii) X is a stack Symbol, that is a member of Γ

The Output of δ is finite set of pairs
(p, γ) where:
p is a new state
γ is a string of stack symbols that replaces X at the top of the Stack

Eg,

If $\gamma = \epsilon$ then the stack is poped

If $\gamma = X$, then the stack is unchanged

If $\gamma = YZ$, then X is replaced by Z and Y is pushed onto the stack

$Q, \Sigma, \Gamma, \delta, q_0, z_0, F$

87 PDA (Graphical Notation)

FSM $A \xrightarrow{a} B$

PDA

$A \xrightarrow{a,\, b \to c} B$

This symbol pushed onto the Stack, ϵ mean nothing is pushed

input symbol may be ϵ

symbol on Top of the Stack. This Symbol is poped

↓

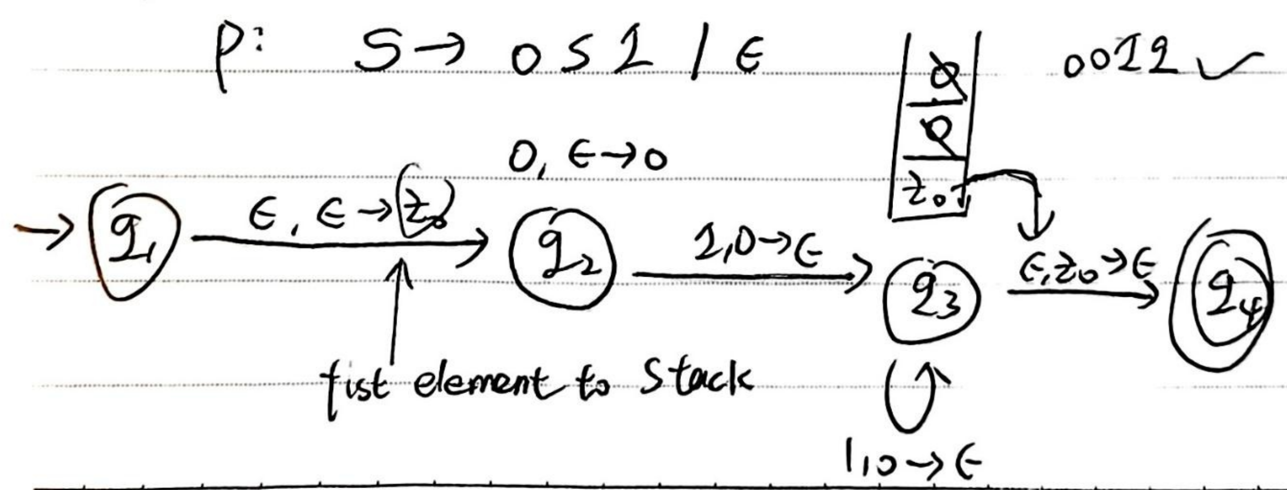
ϵ means the stack is neither read nor poped

)

$CFG = \{V, T, S, P\}$

Example: Construct a PDA $L = \{0^n 1^n \mid n \geq 0\}$

P: $S \to 0S1 \mid \epsilon$



0011 ~~push → pop~~ | pop → push

this a non-deterministic ~~PSD~~ PDA

(86) Pushdown- Automata - Example

$L=\{ww^R \mid w=(a+b)^+\}$

~~$S \to aS \mid bS \mid SS$?~~

PDA Accepts Even Palindromes:

$L=\{ww^R \mid w=(a+b)^+\}$

Example: NOON

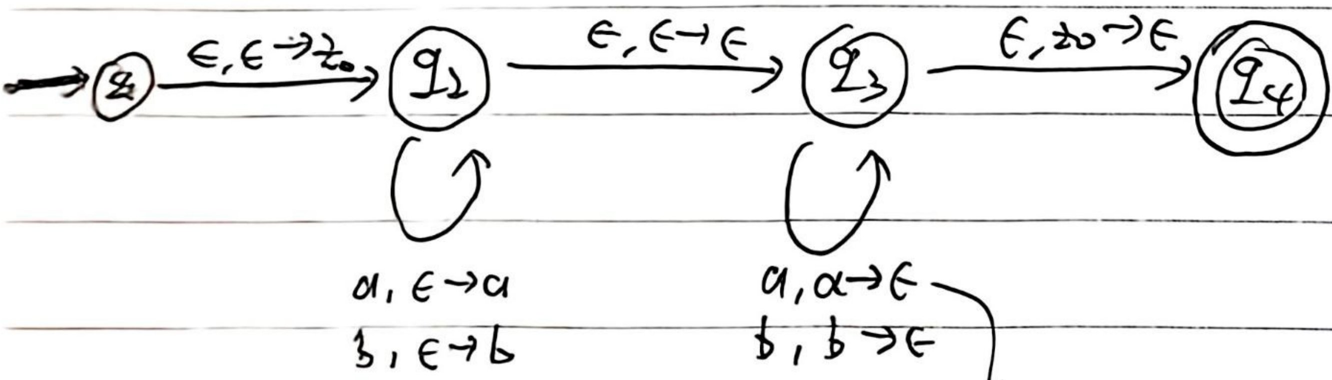
NO LEMON NO MELON

123321

abba

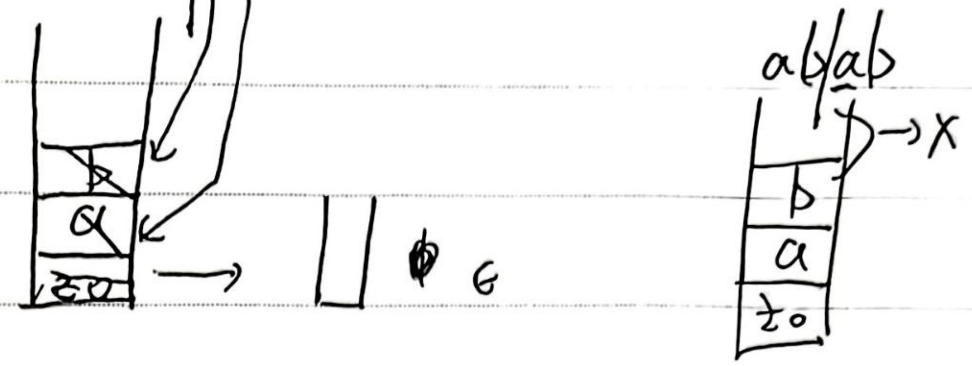
RACECAR

at middle of input string



if a is on the top of stack

ex abba ✓



how do I know the input string have arrived the middle?

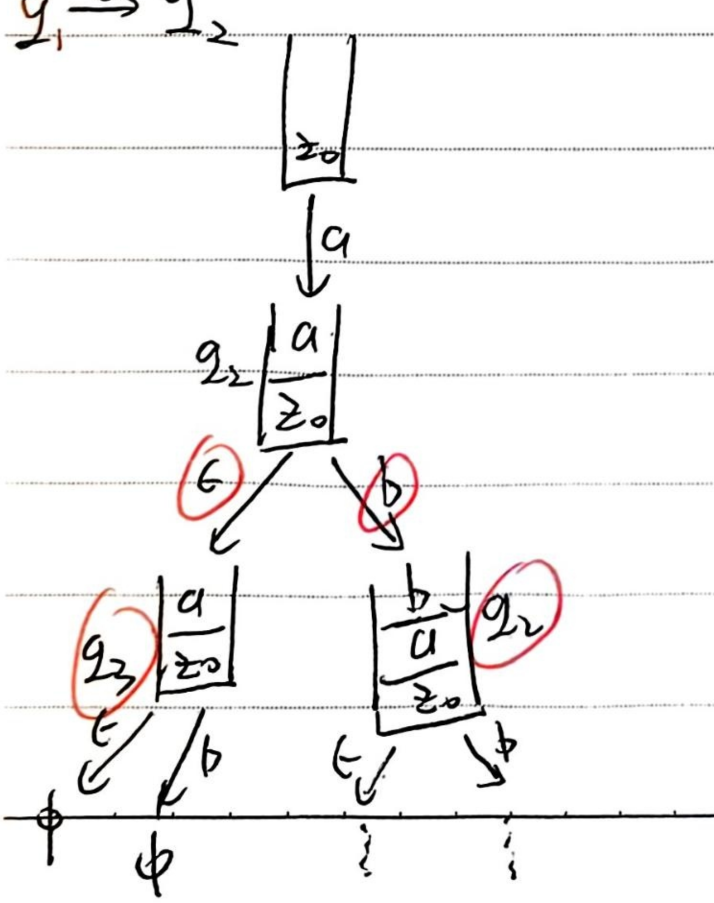
89

the answer is no, can't know,

⇒ assume

"εaεbεbεaε"

$q_1 \xrightarrow{\epsilon} q_2$



→ abεba

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90 Example 2,

$\epsilon a \epsilon b \epsilon a \epsilon b$